AI-Based Multimodal Fake Review Detection System Using Natural Language Processing and Behavioural Analysis

Improving E-Commerce Trust Through Intelligent Fake Review Detection

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***Abstract* —** The rapid expansion of e-commerce platforms has significantly increased the influence of online customer reviews on purchasing decisions. However, the growing prevalence of fake and deceptive reviews has become a major concern, affecting customer trust, business reputation, and platform credibility. Existing fake review detection techniques primarily focus on textual analysis and often fail to identify sophisticated fraudulent reviews that mimic genuine customer experiences.

***Keywords* —** Fake Review Detection, Natural Language Processing, Machine Learning, Sentiment Analysis, Random Forest, TF-IDF, Behaviour Analysis, E-Commerce Security

INTRODUCTION

Online reviews have become one of the most influential factors affecting customer purchasing decisions in modern e-commerce platforms such as Amazon, Flipkart, Yelp, and Alibaba. Customers frequently rely on reviews to assess product quality, compare alternatives, and reduce uncertainty before making purchases. Consequently, reviews significantly impact product popularity, revenue generation, and brand reputation.

Despite their importance, online review systems have become vulnerable to manipulation through fake reviews. Fraudulent reviews are intentionally created to mislead customers by artificially increasing product ratings or harming competing products. Such reviews can distort consumer perceptions and negatively affect the reliability of digital marketplaces. Traditional fake review detection methods mainly depend on sentiment analysis and keyword-based filtering mechanisms. However, these approaches often fail when deceptive reviews are written using natural and convincing language.

To address this challenge, this research proposes a multimodal fake review detection framework that analyses both review text and reviewer behaviour. By combining NLP-based textual analysis with behavioural features such as review frequency, rating consistency, and reviewer activity, the proposed system achieves more reliable fake review identification.

LITERATURE REVIEW

Several studies have explored the problem of fake review detection using different techniques.

Author	Technique	Contribution	Limitation
Jindal & Liu	Opinion Spam Detection	Introduced Fake Review Detection	Limited Features
Ott et al.	NLP	Linguistic Analysis	Small Dataset
Mukherjee et al.	Behaviour Analysis	Reviewer Group Detection	Needs User History
Ren & Ji	LSTM	Better Accuracy	High Cost
Alghamdi et al.	Survey	Research Directions	Dataset Limitations

Table 01. Literature Review and Comparative Analysis of Existing Techniques

PROBLEM STATEMENT

Online review systems are increasingly affected by fake, spam, and misleading reviews that influence customer purchasing decisions and reduce platform credibility. Existing review filtering mechanisms often fail to detect sophisticated deceptive reviews because they rely primarily on textual features.

The main challenges include:

- Detecting reviews written in human-like language.
- Identifying abnormal reviewer behaviours.
- Managing large volumes of review data.
- Improving overall classification accuracy.

Therefore, there is a requirement for an intelligent review analysis system capable of utilizing both textual content and reviewer behavioural information to distinguish genuine reviews from fraudulent ones.

PROPOSED METHODOLOGY

The proposed fake review detection system consists of five major phases.

A. Data Collection:

Review data is collected from publicly available datasets such as Amazon Review Dataset, Yelp Open Dataset, and Kaggle Fake Review Detection Dataset.

The collected dataset includes review text, user information, ratings, timestamps, and review labels.

Dataset Statistics -

Parameter	Value
Total Reviews	12,000
Genuine Reviews	6,000
Fake Reviews	6,000
Training Dataset (80%)	9,600
Testing Dataset (20%)	2,400

Table 02. Dataset Statistics of Genuine and Fake Reviews

Dataset Attributes -

Attribute	Description
Review Text	Customer review content
Rating	Product rating (1-5)
User ID	Reviewer identifier
Timestamp	Review submission date
Review Length	Number of words
Sentiment Score	Polarity score
Label	Genuine / Fake

Table 03. Dataset Attributes Description

B. Data Preprocessing:

Before model training, the collected review dataset undergoes preprocessing which includes:

- Lowercase conversion
- Removal of punctuation and special characters
- URL removal
- Tokenization

These processes convert unstructured review text into a machine-readable format suitable for feature extraction and classification.

C. Feature Extraction:

Textual Features -

- TF-IDF Scores, Word Frequency, Review Length, Sentiment Polarity, N-Gram Features

Behavioural Features -

- Review Frequency, Rating Consistency, Reviewer Activity, Temporal Posting Patterns, Account Behaviour Indicators

The combination of these features allows the model to identify hidden patterns associated with fake reviews.

D. Machine Learning Models:

The following algorithms are implemented & compared:

- Logistic Regression, Naïve Bayes, Support Vector Machine (SVM), Random Forest
- Each model is trained using extracted features and evaluated using standard classification metrics.

MATHEMATICAL FORMULA'S

TF-IDF -> $TF\text{-}IDF(t,d) = TF(t,d) \times \log(N/DF(t))$

Where:

- TF = Term Frequency
- DF = Document Frequency
- N = Total Documents

Accuracy ->

$Accuracy = (TP + TN) / (TP + TN + FP + FN)$

Precision -> $Precision = TP / (TP + FP)$

Recall -> $Recall = TP / (TP + FN)$

F1 Score ->

$F1 = 2 \times (Precision \times Recall) / (Precision + Recall)$

GRAPHICAL VIEW

Data Distribution Pie Chart -

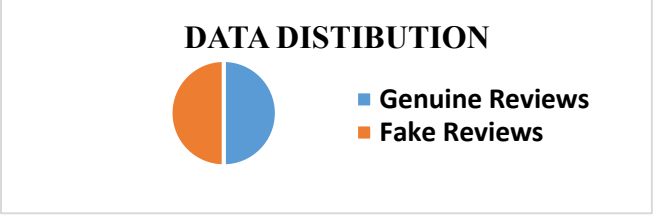


Fig 01. Dataset Distribution of Genuine and Fake Reviews

Feature Importance Graph –

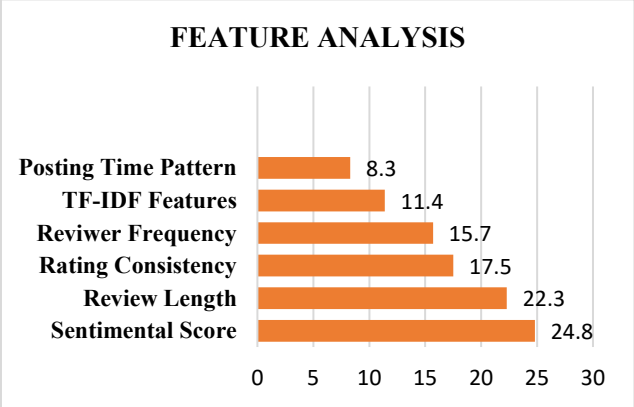


Fig 02. Feature Extraction Process Using NLP and Behavioural Features

Confusion Matrix Heatmap –

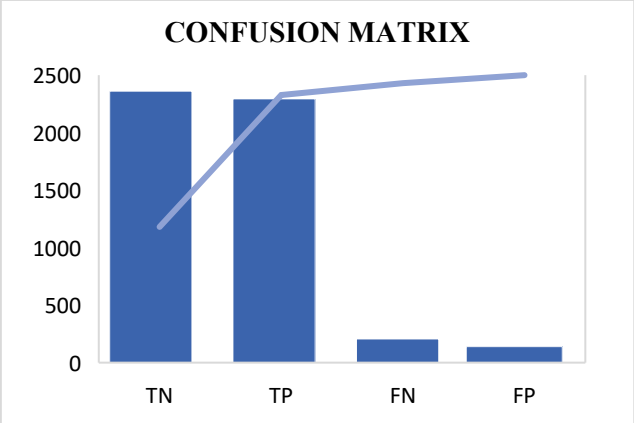


Fig 03. Confusion Matrix Heatmap of Random Forest Classifier

SYSTEM ARCHITECTURE

The architecture ensures scalability, modularity, and efficient processing of reviews from multiple e-commerce platforms.

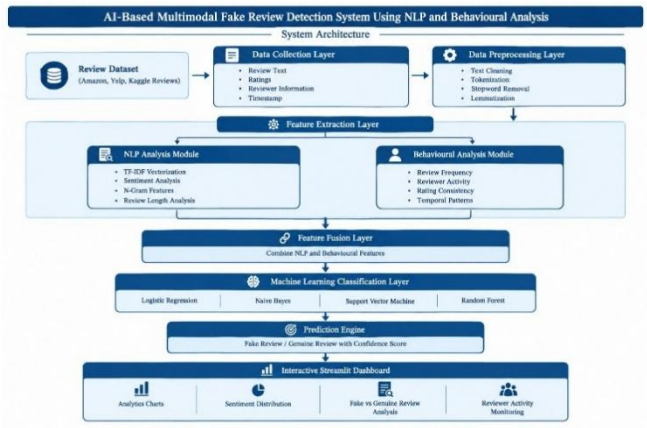


Fig 04. Proposed System Architecture

IMPLEMENTATION

The system is implemented using Python and Streamlit.

The GUI allows users to enter a review and instantly determine whether it is genuine or fake.

Component	Technology
Programming Language	Python 3.11
Frontend	Streamlit
NLP Libraries	NLTK, SpaCy
ML Library	Scikit-Learn
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn, Plotly
Database	CSV / SQLite

Table 04. Technologies and Tools Used in System Development

PERFORMANCE EVALUATION

To evaluate classification performance, four metrics are used.

- **Accuracy-** Accuracy measures the overall correctness of the model.
- **Precision-** Precision measures the proportion of correctly identified fake reviews among all reviews classified as fake.
- **Recall-** Recall measures the proportion of actual fake reviews successfully detected by the model.
- **F1-Score-** F1-Score provides a balanced measure between precision and recall.

Model	Accuracy	Precision	Recall	F1 Score
Naïve Bayes	84.6%	83.4%	82.5%	82.9%
Logistic Regression	88.2%	87.6%	86.9%	87.2%
SVM	91.3%	90.7%	90.2%	90.4%
Random Forest	93.1%	92.4%	91.8%	92.1%

Table 05. Performance Comparison of Machine Learning Models

The Random Forest classifier achieved the highest performance among all evaluated machine learning algorithms. The ensemble learning approach effectively

CONFUSION MATRIX ANALYSIS

Actual / Predicted	Genuine	Fake
Genuine	2360	140
Fake	205	2295

Table 06. Confusion Matrix of Random Forest Classifier

The confusion matrix indicates that the Random Forest model correctly classifies most genuine and fake reviews with minimal classification errors.

DASHBOARD ANALYSIS

The developed Streamlit dashboard provides:

- Fake vs Genuine Review Distribution
- Rating Distribution Analysis
- Sentiment Analysis Visualization
- Word Frequency Analysis
- Reviewer Activity Monitoring

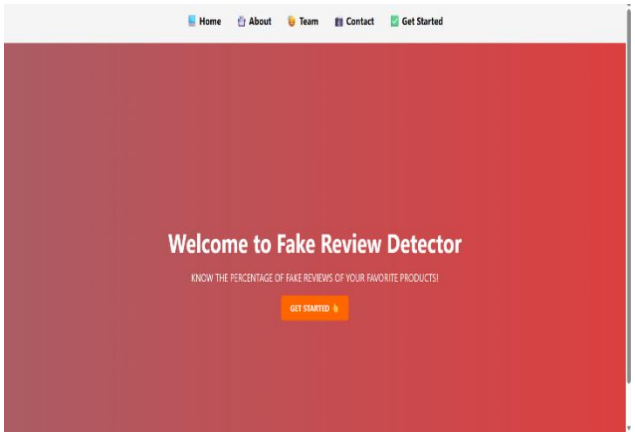


Fig 05. Home Interface of the System

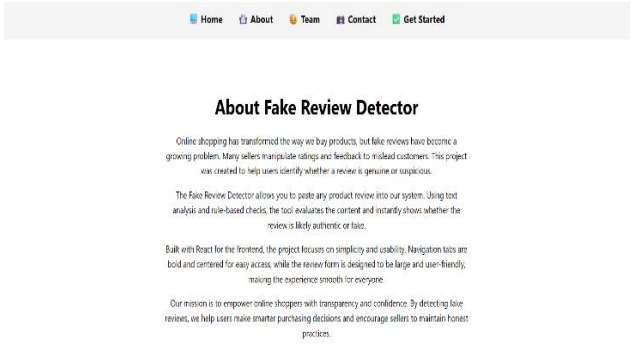


Fig 06. About Page of the System

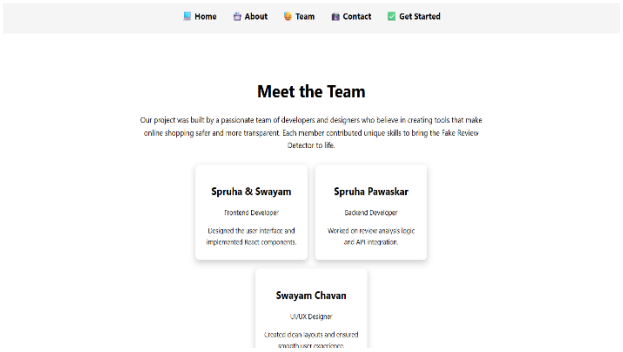


Fig 07. Team Page of the System

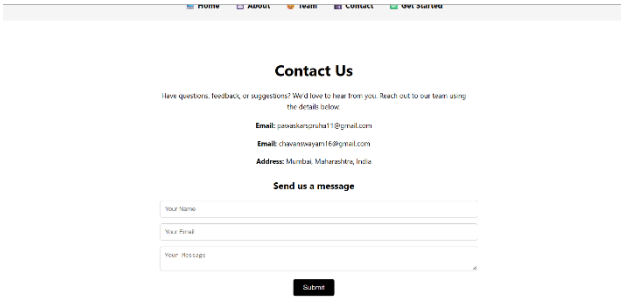


Fig 08. Contact Page of the System

These visualizations assist administrators in understanding review patterns and detecting suspicious activities.

ADVANTAGES OF THE PROPOSED SYSTEM

- Improved fake review detection accuracy.
- Hybrid NLP and behavioural analysis approach.
- Support for large review datasets.
- Interactive dashboard visualization.
- Real-time prediction capability.
- Reduced review manipulation and fraud.

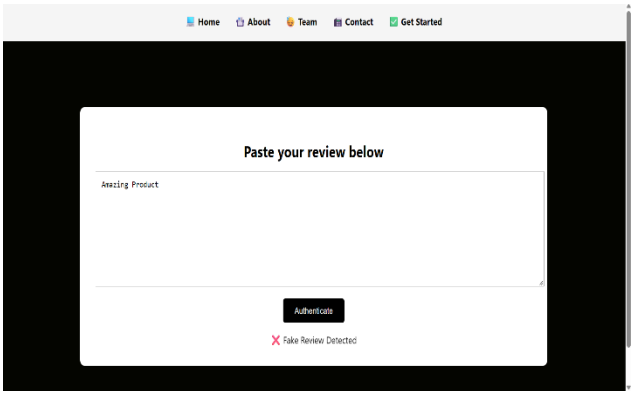


Fig 09. Fake Review Prediction Output

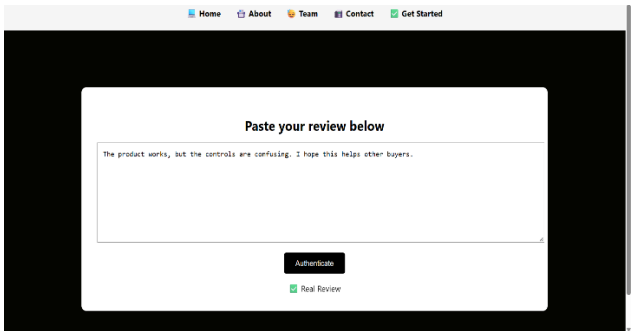


Fig 10. Genuine Review Prediction Output

FUTURE SCOPE

The system can be enhanced by incorporating:

- BERT and Transformer-based models.
- Deep Learning techniques such as LSTM and GRU.
- Real-time review monitoring.
- Multilingual fake review detection.
- Graph Neural Networks for reviewer relationship analysis.
- Cloud-based deployment.

Limitations:

- Dataset collected from limited domains.
- Behavioural features may be unavailable for new users.
- English-language reviews only.
- Performance may vary across platforms.

CONCLUSION

This research presents an AI-based multimodal fake review detection system that combines Natural Language Processing and behavioural analysis techniques for accurately identifying deceptive reviews. The proposed framework utilizes textual, sentiment, and behavioural features to distinguish fake reviews from genuine customer feedback.

Experimental comparison of multiple machine learning algorithms demonstrates that Random Forest provides superior classification performance. The developed system contributes to improving trust, transparency, and reliability within online review ecosystems and can serve as a foundation for future intelligent review verification systems.

The proposed framework can assist e-commerce platforms, businesses, and consumers by improving review authenticity, reducing fraudulent activities, and supporting informed decision-making. Therefore, the system provides a scalable foundation for future intelligent review verification and fraud detection applications.

COMPARATIVE ANALYSIS

Feature	Traditional Methods	Proposed System
Text Analysis	Yes	Yes
Sentiment Analysis	Limited	Advanced
Behaviour Analysis	No	Yes
Multiple ML Models	Limited	Yes
Dashboard Support	No	Yes
Detection Accuracy	Moderate	High

Table 07. Comparative Analysis of Existing and Proposed Systems

The proposed system outperforms traditional text-based fake review detection approaches by incorporating reviewer behavioural patterns alongside NLP-based textual analysis.

This hybrid methodology enhances classification accuracy and reduces false positive detections.

APPLICATIONS

The proposed fake review system can be utilized in:

- E-commerce platforms
- Hotel booking systems
- Restaurant review websites
- Online marketplaces
- Product recommendation systems
- Customer feedback analysis systems

NOVELTY

Unlike existing approaches that focus solely on textual analysis, the proposed framework combines textual and behavioural features to improve detection accuracy.

The integration of reviewer activity patterns and sentiment-based linguistic analysis enables more robust identification of deceptive reviews.

ACKNOWLEDGEMENT

The authors would like to express their sincere gratitude to Dr. Ujwala M. Sav for valuable guidance and support throughout the development of this project.

The authors also thank the Department of Information Technology & Data Science, Vidyalankar School of Information Technology, for providing the necessary resources and encouragement to complete this research work successfully.

THREAT TO VALIDITY

The performance of the proposed system depends on the quality and diversity of the training dataset. Results may vary when applied to reviews from different domains or platforms.

Furthermore, reviewer behavioural features may not always be available for newly created accounts. Future studies should utilize larger datasets from multiple platforms to improve generalization.

ETHICAL CONSIDERATIONS

The proposed system is intended to improve transparency and trust within online review systems. Care should be taken to minimize false-positive detections, as incorrect classification may affect legitimate users and businesses.

The system should be used as a decision-support tool rather than a final judgment mechanism.

PRACTICAL IMPLICATIONS

The proposed fake review detection system can assist e-commerce platforms in maintaining review authenticity, improving customer trust, and reducing fraudulent activities.

Businesses can use the system to monitor suspicious reviewer behaviour and ensure fair competition within online marketplaces.

DECLARATION OF ORIGINALITY

The authors hereby declare that this research work is original and has been carried out independently.

All sources of information used in this study have been properly acknowledged and cited. To the best of our knowledge, the work has not been submitted or published elsewhere.

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